BUSINESS CASE STUDY ANALYSIS

Performance Review

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Analysis period: February 2022 – April 2024

Data sources: Business (Transaction Value) & Orders

Calculations performed with python on business.ipynb

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Executive Summary

Over the 27-month period analyzed (February 2022 – April 2024), the business generated a combined Total Transaction Value (TV) across 23 industries, tracked alongside order volume to understand both revenue scale and transaction efficiency. Five findings stand out:

- **Revenue is concentrated in three industries.** Restaurant, Take-out & Catering, E-commerce, and Transportation & Logistic Service together generated $34,497,430 — 44.29% of total revenue. A wider Pareto view shows the top 8 industries (35% of the 23) account for ~80.7% of all revenue.
- **Transaction value and order volume tell different stories.** The industry with the highest revenue (Restaurant, Take-out & Catering) has the lowest Average Order Value ($102.75) — its performance is volume-driven, not order-size-driven. Software & Services, Household & Personal Products, and Insurance operate the opposite way: fewer transactions, much higher value each.
- **Growth is real but uneven.** Retail & Wholesale (R^2=0.921) and Health Care Equipment & Services (R^2=0.930) show strong, statistically reliable upward trends. Household & Personal Products, by contrast, shows essentially no trend (R^2=0.087) — its early spike (June 2022) appears to be a one-time event, not sustained growth.
- **Restaurant, Take-out & Catering is seasonal, not linear.** A linear model poorly explains this industry (R^2=0.38). Residual analysis reveals a recurring December peak and a recurring spring/April dip — a seasonal pattern that a straight-line trend obscures and that has direct implications for inventory planning.
- **Two accounts represent significant concentration risk.** ID 6301097 account for 54.21% of the entire Transportation & Logistic Service industry, and ID 6302851 account for 80.09% of Construction, Machinery & Infrastructure. Combined, these two accounts ($10,824,534) generate more revenue than the entire Media & Entertainment industry ($901,190).

1. Business Performance Overview

The table below ranks all 23 industries by Total Transaction Value (TV) — the sum of all recorded transactions per industry over the 27-month period — alongside the % contribution.

Industry	Total TV	Contribution %
1. Restaurant, Take-out & Catering	$13,785,802	17.7
2. E-commerce	$10,648,449	13.67
3. Transportation & Logistic Service	$10,063,179	12.92
4. Food & Staples Retailing	$6,794,471	8.72
5. Construction, Machinery & Infrastructure	$6,704,979	8.61
6. Retail & Wholesale	$6,501,325	8.35
7. Household & Personal Products	$4,338,574	5.57
8. Health Care Equipment & Services	$4,018,697	5.16
9. Automobiles & Components	$2,568,324	3.30
10. Consumer Durables & Apparel	$2,373,549	3.05
11. Materials	$1,985,906	2.55
12. Professional Services	$1,935,250	2.48
13. Consumer Services	$1,615,550	2.07
14. Technology Hardware & Equipment	$1,567,870	2.01
15. Pharmaceuticals, Biotechnology & Life Sciences	$1,183,515	1.52
16. Media & Entertainment	$901,190	1.16
17. Software & Services	$427,568	0.55
18. Real Estate	$140,278	0.18
19. Insurance	$131,904	0.17
20. Finance	$74,890	0.10
21. Utilities	$53,283	0.07
22. Telecommunication Services	$41,780	0.05
23. Energy	$20,991	0.03

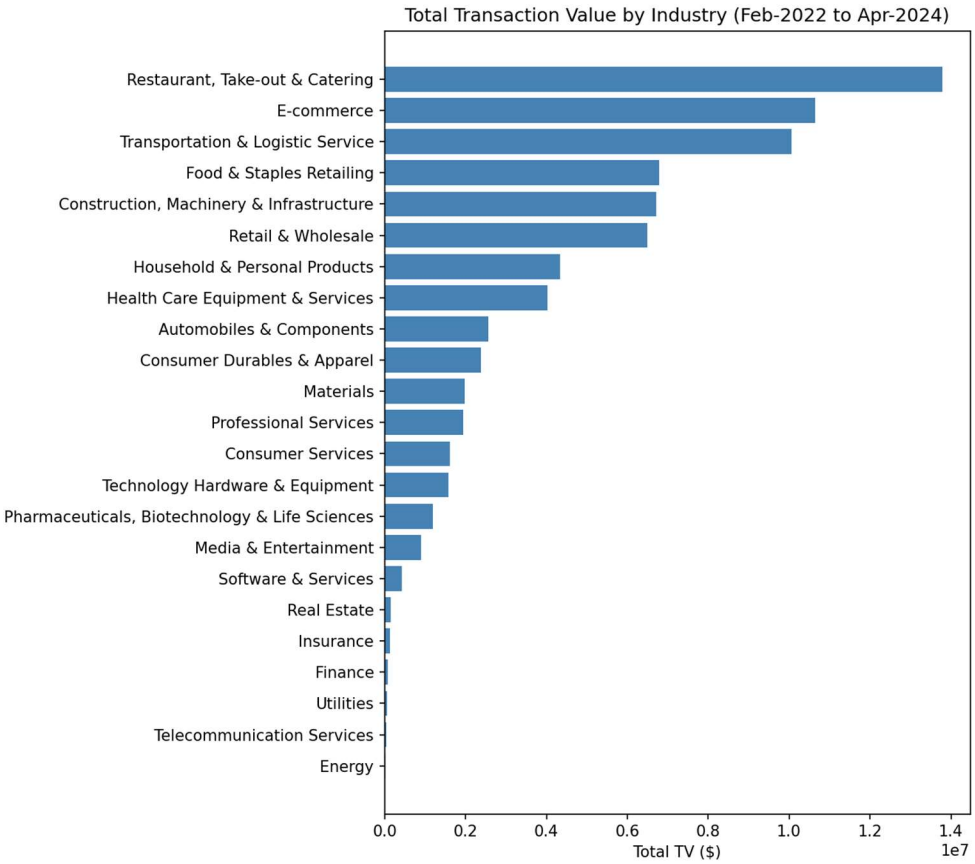


Figure 1. Total Transaction Value by industry, February 2022 – April 2024.

Top 3 Industries

Restaurant, Take-out & Catering, E-commerce, and Transportation & Logistic Service are the three highest-revenue industries, combining for $34,497,430 over the period.

Pareto (80/20) Analysis

Ranking industries by cumulative contribution shows that revenue is not evenly spread across all 23 industries. It takes the top 8 industries (35% of all industries) to reach approximately 80% of total revenue — a moderate concentration, not a strict 80/20 split.

Figure 2. Cumulative revenue contribution by industry, with the 80% threshold marked.

2. Order Performance & Average Order Value

Orders by Industry

Total order counts per industry, over the same period as the revenue analysis above (through April 2024).

Industry	Total Orders
1. Restaurant, Take-out & Catering	134,155
2. E-commerce	72,048
3. Transportation & Logistic Service	54,271
4. Food & Staples Retailing	52,786
5. Retail & Wholesale	43,387
6. Construction, Machinery & Infrastructure	38,124
7. Health Care Equipment & Services	25,010
8. Household & Personal Products	22,754
9. Automobiles & Components	18,224
10. Consumer Durables & Apparel	15,301
11. Professional Services	13,906
12. Materials	12,910
13. Technology Hardware & Equipment	12,834
14. Consumer Services	10,265
15. Pharmaceuticals, Biotechnology & Life Sciences	8,777
16. Media & Entertainment	6,881
17. Software & Services	1,826
18. Real Estate	1,110
19. Insurance	896
20. Finance	576
21. Utilities	379
22. Telecommunication Services	215
23. Energy	131

Average Order Value (AOV)

AOV = Transaction Value ÷ Orders. It measures the average amount spent per transaction, and is calculated here by summing revenue and orders at the industry-month level before dividing (rather than averaging individual ratios), which avoids distortion from small accounts.

Industry	AOV	AOV Max	AOV Min
1. Software & Services	$193.10	$335.28	$61.00
2. Household & Personal Products	$187.05	$727.76	$142.16
3. Insurance	$186.80	$436.80	$98.80
4. Transportation & Logistic Service	$185.12	$349.68	$96.90
5. Construction, Machinery & Infrastructure	$176.99	$227.23	$140.59
6. Health Care Equipment & Services	$169.41	$219.21	$127.21
7. Materials	$161.65	$231.32	$116.84
8. Telecommunication Services	$157.26	$281.07	$76.90
9. Consumer Services	$153.80	$186.60	$124.49
10. Consumer Durables & Apparel	$149.54	$191.19	$107.29
11. E-commerce	$147.82	$185.13	$126.91
12. Retail & Wholesale	$146.40	$183.03	$127.45
13. Automobiles & Components	$141.42	$171.04	$124.90
14. Energy	$140.46	$256.84	$51.00
15. Professional Services	$140.13	$178.08	$125.02
16. Pharmaceuticals, Biotechnology & Life Sciences	$139.40	$189.95	$119.68
17. Food & Staples Retailing	$126.29	$158.25	$95.28
18. Utilities	$125.01	$394.24	$40.33
19. Media & Entertainment	$121.33	$169.89	$80.10
20. Real Estate	$119.69	$174.21	$69.38
21. Finance	$119.30	$174.70	$63.19
22. Technology Hardware & Equipment	$113.23	$180.73	$59.68
23. Restaurant, Take-out & Catering	$102.75	$137.41	$81.01

Key insight: the industry with the highest total revenue, Restaurant, Take-out & Catering, has the **lowest AOV of all 23 industries ($102.75)**. Its revenue is driven by high transaction volume, not high-value transactions — the opposite pattern from Software & Services, Household & Personal Products, and Insurance, which post the three highest AOVs and depend on fewer, larger transactions.

3. Trend Analysis

A linear regression model was fit to each industry's monthly Transaction Value to identify underlying growth trends, independent of month-to-month noise. The model takes the form y = ax + b, where *x* is the sequential month number and *y* is the predicted transaction value; R^2 indicates how well the line explains the actual data (1.0 = perfect fit, 0 = no explanatory power).

Strongest Growth Trends

Retail & Wholesale shows the strongest, most reliable growth in the dataset: transaction value rose from approximately $25,600 to $529,500 over the period — roughly a 20-fold increase — with a linear fit that explains 92.1% of the variance (R^2 = 0.921).

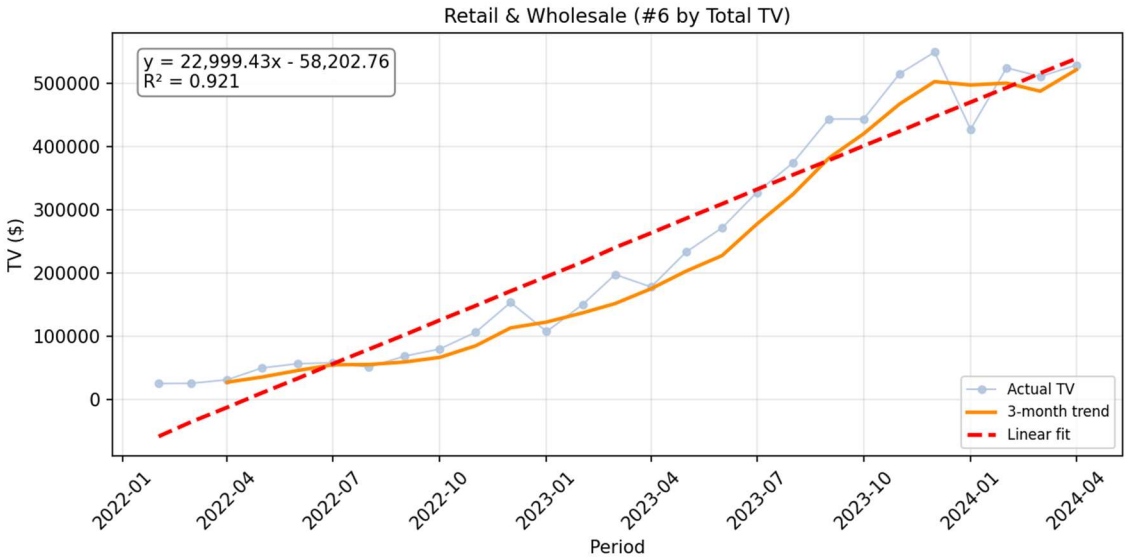


Figure 3. Retail & Wholesale: actual transaction value, 3-month trend, and linear fit.

Health Care Equipment & Services shows a similarly strong and reliable trend (R^2 = 0.930), growing from approximately $38,700 to $261,200 — roughly a 6.7-fold increase over the same period.

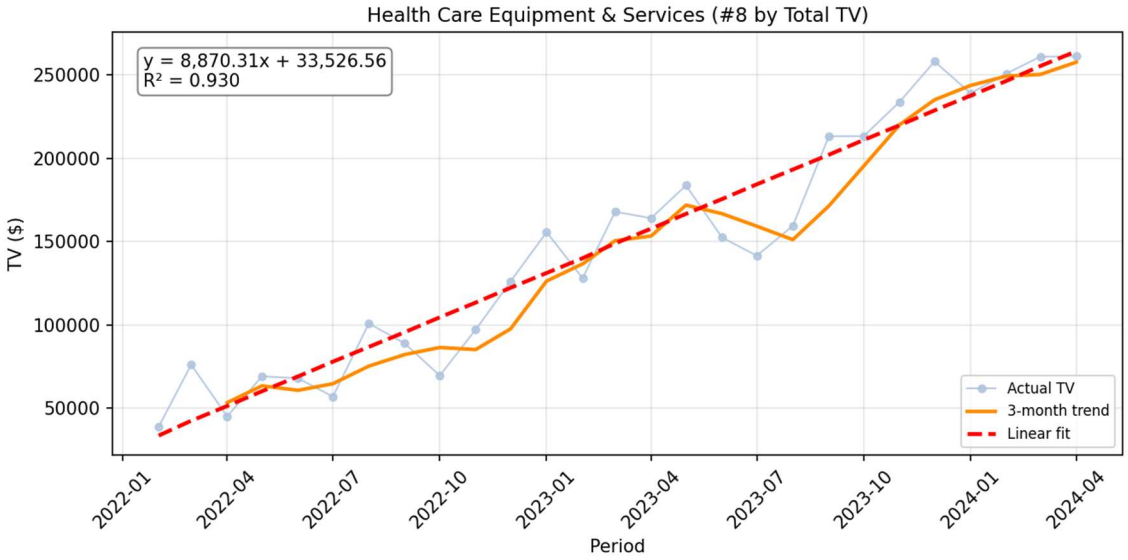


Figure 4. Health Care Equipment & Services: actual transaction value, 3-month trend, and linear fit.

Both industries show accelerating, statistically well-supported growth and are strong candidates for scalable operational investment (see Action Items).

Non-Growth Trend

Household & Personal Products tells a different story. The linear fit is very weak ($R^2 = 0.087$) and its slope is slightly negative, meaning there is no reliable growth trend across the period.

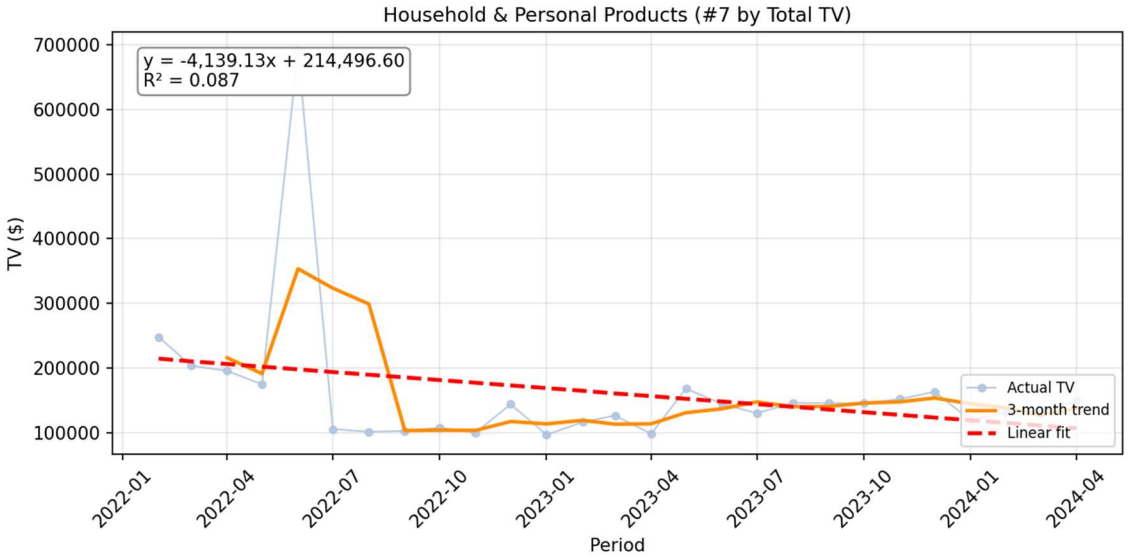


Figure 5. Household & Personal Products: a single spike (June 2022) dominates the series; the rest of the period is essentially flat.

The apparent early strength in this industry is driven almost entirely by a single outlier month — June 2022, where transaction value reached $689,918, more than 3x the industry's typical monthly level. Excluding that one month, the series is flat to slightly declining. This is a "one-hit wonder," not a growth pattern, and warrants root-cause investigation (see Action Items).

High-Volume, Low-Fit Industry: A Case for Seasonality

Restaurant, Take-out & Catering — the single highest-revenue industry — does not fit a linear trend well ($R^2 = 0.38$). Its data points spread widely around the fitted line, suggesting the underlying pattern is seasonal rather than a steady linear trend.

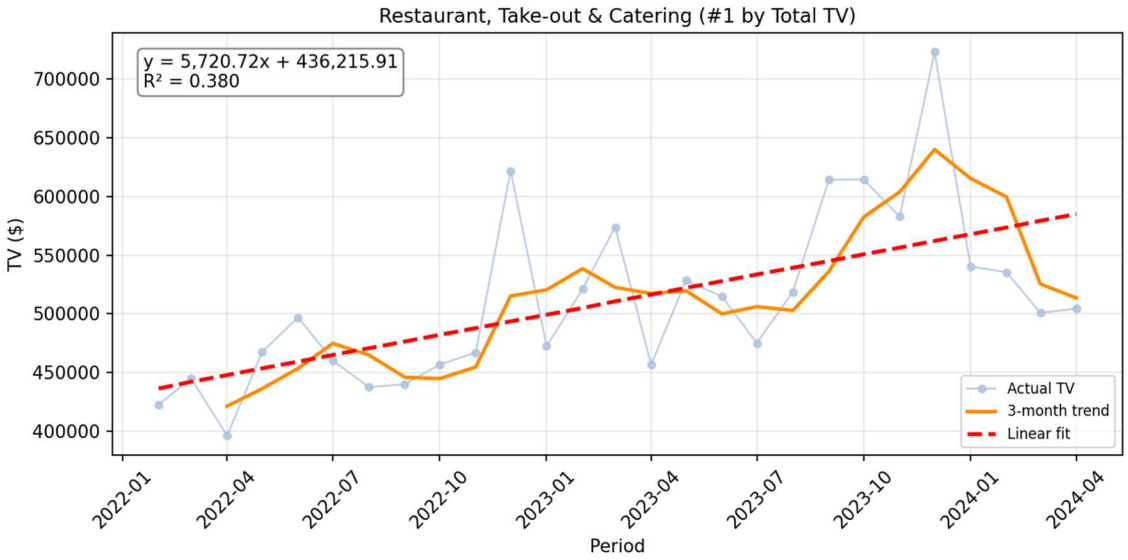


Figure 6. Restaurant, Take-out & Catering: the linear fit poorly explains month-to-month variation.

To separate genuine spikes/dips from random noise, we calculated the residual (actual value minus the value predicted by the line) for every month, and flagged any month more than one standard deviation above or below the trend as a peak or valley, respectively.

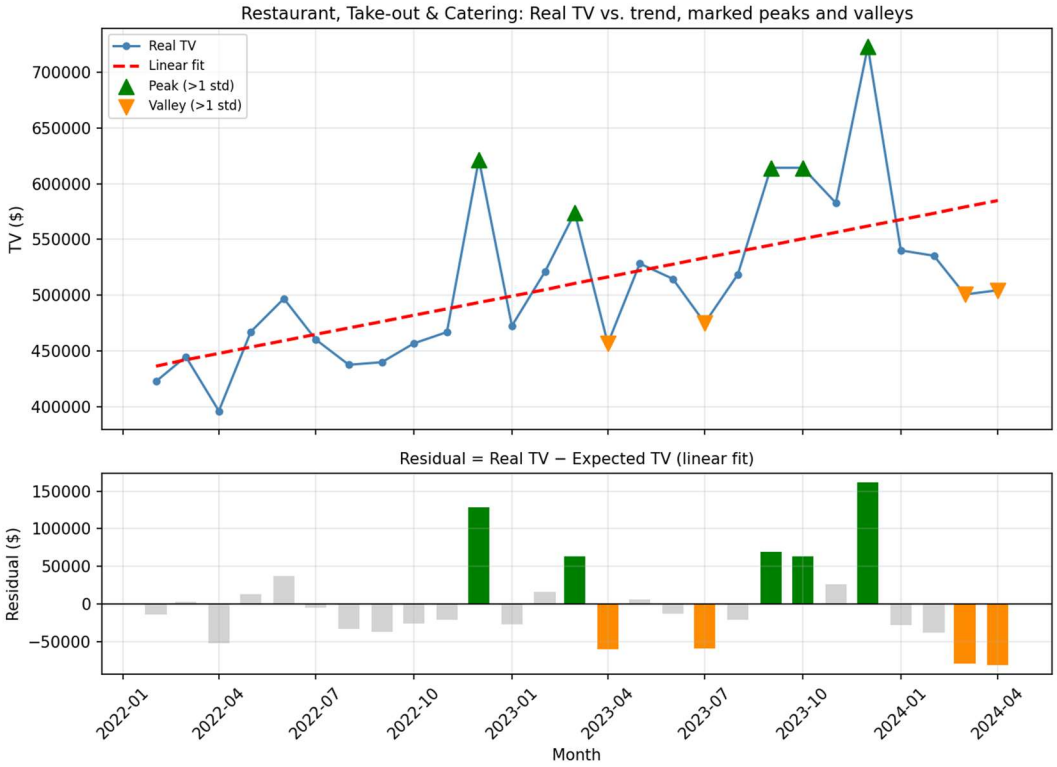


Figure 7. Residual analysis: peaks and valleys relative to the linear trend.

Peak Month	Real TV	Expected TV	Residual
2023-12	$723,188	$562,072	+$161,116
2022-12	$621,841	$493,423	+$128,418
2023-09	$614,371	$544,910	+$69,461
2023-10	$614,371	$550,630	+$63,740
2023-03	$574,018	$510,585	+$63,433

Valley Month	Real TV	Expected TV	Residual
2024-04	$504,381	$584,955	-$80,574
2024-03	$500,617	$579,234	-$78,617
2023-04	$456,536	$516,306	-$59,770
2023-07	$475,011	$533,468	-$58,457

Pattern identified: both December months in the dataset (2022 and 2023) register as peaks, and both April months (2023, 2024) register as valleys — a consistent seasonal pattern across the two years observed. March is inconsistent between years and does not show a clear seasonal signal. With only two years of data, this pattern is a strong preliminary signal, not yet statistically confirmed — a third full year of data would substantially increase confidence.

4. Account (ID) Concentration Analysis

Each industry is composed of many individual accounts (IDs). To assess concentration risk, we summed Transaction Value per account within each industry and calculated each account's share of its industry total. Two industries show extreme concentration in a single account(ID).

Transportation & Logistic Service

84 distinct accounts generated $10,063,179 in total industry revenue. **A single account, ID 6301097, represents 54.21% of the entire industry** ($5,454,820).

ID	Total TV	Contribution %
6301097	$5,454,820	54.21%
6306244	$1,182,452	11.75%
6302863	$819,587	8.14%
6303563	$310,078	3.08%
6302969	$277,367	2.76%

Construction, Machinery & Infrastructure

121 distinct accounts generated $6,704,979 in total industry revenue. **A single account, ID 6302851, represents 80.09% of the entire industry** ($5,369,715).

ID	Total TV	Contribution %
6302851	$5,369,715	80.09%
6303872	$207,836	3.10%
6301475	$123,184	1.84%
6303785	$109,791	1.64%
6302800	$62,023	0.93%

Combined impact: together, ID 6301097 and ID 6302851 generate $10,824,534 — more than the entire Media & Entertainment industry ($901,190), roughly 12 times over. Losing either account would materially impact its respective industry's reported performance; both merit dedicated account management (see Action Items).

5. Action Items

- **Investigate Household & Personal Products.** Since this industry shows no reliable growth trend ($R^2 = 0.087$) and its early strength was driven by a single outlier month (June 2022), conduct a root-cause analysis to understand why the industry is not showing sustained growth like its peers.
- **Plan inventory and staffing around seasonality.** Restaurant, Take-out & Catering shows a consistent December revenue spike and an April dip — align inventory, staffing, and promotional planning with this seasonal pattern.
- **Formalize engagement with concentrated accounts.** ID 6301097 (54.21% of Transportation & Logistic Service) and ID 6302851 (80.09% of Construction, Machinery & Infrastructure) each represent single points of failure for their industries. Define dedicated account ownership and consider 1–3 year SLAs or contracts to reduce churn risk.
- **Develop retention strategies for high-AOV industries.** Software & Services, Household & Personal Products, and Insurance post the highest Average Order Values — retaining these fewer, higher-value transactions has outsized revenue impact per account.
- **Scale operations behind proven growth.** Retail & Wholesale and Health Care Equipment & Services show the strongest, most statistically reliable growth trends ($R^2 > 0.92$) — prioritize scalable processes, automation, and formal KPI tracking to sustain this momentum.